Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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Abstract

Background/Aims: In spatial cluster randomized trials, treatment effects can spill over from treated clusters to neighboring clusters, so the choice of which clusters to treat affects how accurately the treatment effect can be estimated. We compared six treatment assignment designs and simple random sampling (SRS) on two criteria: average estimation error and the error under an unfavorable realized assignment (allocation risk). We then applied the designs to planning a sudden unexpected death (SUD) prevention education pilot in North Carolina.

Methods: We simulated trials on a 10×10 grid of 100 clusters, varying incidence heterogeneity, spatial dependence, spillover magnitude, spillover regime (control clusters only or both arms) and treatment-effect size. The treatment effect was estimated with a maximum likelihood spatial autoregressive (spatial lag) model, and accuracy was measured by mean squared error (MSE). A focused pilot simulated repeated outcomes for each sampled assignment to estimate how MSE varies across assignments. The application used observed 2018–2021 SUD incidence in 58 North Carolina community-college service-area clusters to guide assignment and simulated continuous education outcomes; incidence determined which clusters were treated but did not enter the outcome model. Queen contiguity was primary, with sensitivity analyses using rook contiguity, a hypothetical incidence-related outcome baseline and alternative regional incidence summaries.

Results: The spillover regime determined which designs performed best. On the grid at a treatment effect of one, incidence-guided saturation had a control-only mean MSE of 0.1355 versus 0.2146 for SRS, while SRS remained competitive when spillover reached both arms. In the North Carolina application, incidence-guided saturation lowered control-only mean MSE by 21.1–29.9% relative to SRS, depending on the year, and its worst-performing assignments also had lower error than those of SRS, although which assignments fell in the worst decile was uncertain. Under both-arms spillover, SRS and several randomized designs had similar accuracy. Application coverage of nominal 95% Wald intervals was 92.6–93.6%; a t-based interval with a degrees-of-freedom-corrected standard error restored 94–96%. All application simulations met prespecified precision targets for simulation error with no estimation failures.

Conclusions: Saturation designs merit consideration when spillover is expected mainly in control clusters, and SRS remains competitive when spillover reaches both arms.

Incidence-guided saturation directs the highest treatment saturation to high-burden regions while maintaining accuracy and limiting allocation risk under control-only spillover, and the redirection grows with regional differences in burden. Design choice should also account for model assumptions, treatment budgets, interval undercoverage and uncertainty in worst-case estimates. This study evaluates trial designs, not the effectiveness of an implemented intervention.

Keywords: cluster randomized trial; spatial spillover; interference; design-stage treatment allocation; incidence-guided design; sudden unexpected death.

Introduction

Cluster randomized trials assign groups rather than individuals when interventions operate through clinics, schools or geographic areas.[1,2] Spatial interference challenges the assumption that a cluster's outcome depends only on its own treatment: adjacent populations can share services, information or environmental exposures.[3,4] An allocation may therefore provide substantially different information about a treatment effect than another allocation with the same number of treated clusters.

Design-stage methods include covariate-constrained randomization, randomized saturation and independent-set or spatial cluster designs.[5–8] They address different objectives and constraints. Balancing measured incidence does not ensure treatment and spillover are distinguishable; maximal interspersion can itself make those covariates dependent. Appropriate spatial analysis remains necessary alongside allocation design.

In previous work, we compared simple random assignment and block-stratified sampling on small grids.[9] That work weighed average estimation accuracy against the risk of a poor realized allocation in the studied settings. We extend that question to heterogeneous incidence, a larger grid, additional strategies and an irregular observed geography. We include both simple random sampling (SRS), meaning complete randomization of a fixed number of clusters, and the strict alternating allocation called Checkerboard, and we evaluate both the mean squared error (MSE) of the treatment-effect estimate and how that error varies across possible assignments. We do not directly compare numerical results between studies.

Our application supports planning an education intervention for healthcare professionals involved in sudden unexpected death (SUD) prevention. Historical incidence can identify areas of need without implying that education ability, or the intervention's effect on it, depends on incidence. We consequently separate incidence-guided allocation from the simulated education baseline. A trial may realize only one assignment, making both expected accuracy and poor-allocation downside relevant.

Incidence can be used to direct resources toward areas with greater historical burden, to balance that burden between arms, or to set local treatment saturation. These strategies express different priorities even with similar treatment budgets (numbers of treated clusters), and they need not provide the same information for separating treatment from neighboring exposure. We

evaluated their estimation consequences under specified spatial models, including whether the design with the lowest average error also protects against an unfavorable realized allocation.

Methods

Simulation study on a 100-cluster grid

We used a 10×10 grid with row-standardized weights W. Queen neighbors share an edge or corner; rook neighbors share an edge. Queen was primary, with rook assessed separately. Outcomes followed

$$Y = (I - \rho W)^{-1}[\tau Z + S(Z) + \beta X + \epsilon], \qquad \epsilon \sim N(0, I), \tag{1}$$

where Z is treatment and τ its structural coefficient before spatial propagation. It is not an average total effect after applying the spatial multiplier. We set $\beta = 1$ in the grid study. Spillover was $S(Z) = \gamma WZ$ for both arms or $S(Z) = \gamma(1 - Z) \circ WZ$ for controls only, with elementwise product $\circ$. The parameter grid crossed $\rho = 0, 0.01, 0.2, 0.5$, $\gamma = 0.5, 0.6, 0.7, 0.8$ and $\tau = 0.8, 1, 1.5, 2, 3$.

The incidence vector X followed five configurations: independent Uniform$(0, 1)$ incidence, spatial incidence at two dependence strengths ($\rho_X = 0.2, 0.5$) and Poisson incidence at the same two strengths. Spatial incidence used a queen-contiguity spatial autoregressive (SAR) filter followed by a normal distribution-function transformation. Poisson counts used spatial log-rates with base rate 35 per 100,000 and equal populations of 100,000; observed rates were rank-scaled to $(r_i - 0.5)/100$, using average ranks for ties. These are general simulation settings, not calibration to the application counts. Each configuration's ten independent surfaces were shared across parameters. For surface k, assignment, outcome generation and fitted incidence all used X_k.

The six designs were: Checkerboard; High Incidence Focus, treatment of the highest-incidence half; Isolation Buffer, a greedy random maximal independent set; 2×2 Blocking, randomization of two clusters in every 2×2 block; Balanced Quartiles, balancing by incidence quartiles; and Incidence-Guided Saturation, which assigned 80/60/40/20% treatment within four quadrants ordered by mean incidence. All treated 50 clusters except Isolation

Buffer, whose budget depended on the graph. SRS completely randomized 50 of 100. Table 1 summarizes the rules and their geographic adaptations; supplementary material gives full specifications. Checkerboard was deterministic; incidence ties were broken randomly for each assignment in ranking-based designs. High Incidence Focus had only one possible assignment when no tie affected selection.

The strategies vary both incidence balance and spatial exposure. Quartile stratification distributes treatment within incidence groups; High Incidence Focus instead aligns treatment with incidence. 2×2 Blocking imposes local treatment balance, while saturation permits intentionally different local proportions. Isolation Buffer prevents treated neighbors, and Checkerboard imposes strong interspersion. These restrictions can affect the relationship between treatment and spillover covariates, beyond the number treated. We also evaluated two related variants, saturation with randomly ordered quadrants and balancing within incidence halves. Each performed similarly to its retained counterpart, so the main text reports the six designs; results for both variants are in the supplementary material.

The full benchmark, including the two variants, contained 14,400 scenarios per estimator, each with ten incidence surfaces and 25 assignment–outcome pairs per surface, giving 250 fits, each with independent outcome noise. Designs compared under the same parameter setting shared surfaces and outcome noise, and the treatment-effect levels reused the same assignment and noise draws. Scenario metrics were MSE, bias and coverage of a 95% t-based confidence interval with a degrees-of-freedom-corrected standard error. Because each metric is estimated from a finite number of simulated datasets, it carries simulation (Monte Carlo) error; its Monte Carlo standard error (MCSE) measures this uncertainty and shrinks as more datasets are simulated. MCSEs and paired comparisons were based on the 50 independent configuration-by-surface units, rather than treating settings that share surfaces as independent replications.

Estimating treatment effect and allocation risk

The primary maximum likelihood (ML) fit of the SAR (spatial lag) model included an intercept, treatment, the true regime-specific spillover covariate and grid incidence. Our concentrated-likelihood implementation matched `spatialreg::lagsarlm` in validation; confidence intervals were $\hat{\tau} \pm 1.96\widehat{\mathrm{SE}}$, with SE the estimated standard error.[10,11] This assumes

the analyst knows the spillover form. A grid sensitivity omitted spillover. We recorded coefficients dropped as non-identifiable (aliased), software warnings, failed estimates or intervals, and the rank of the design matrix. Rook Checkerboard has $WZ = 1 - Z$, so treatment and spillover are not separately identified; its retained treatment coefficient estimates $\tau - \gamma$. For queen Checkerboard under control-only spillover, identification relies heavily on clusters at the grid boundary. The rook estimate does not recover the target effect; queen Checkerboard remains identified but can be imprecise.

For assignment a, allocation risk is $m(a) = E_\epsilon[(\hat{\tau} - \tau)^2 \mid a, X]$. Simulating repeated outcomes for each sampled assignment estimates $m(a)$; across assignments we then report the 90th percentile of $m(a)$ (q90) and the mean of the worst 10% of sampled assignments (worst-decile mean). These differ from quantiles of realized errors or scenario-level MSE. A focused grid pilot used two fixed surfaces per configuration, queen weights, the four combinations of $\rho \in \{0, 0.5\}$ and $\gamma \in \{0.5, 0.8\}$, 100 assignment draws and 100 outcomes per distinct assignment, increasing outcomes to 400 for selected assignments. Its findings are supporting pilot evidence, separate from the full grid comparison.

Repeated outcomes are essential to interpreting downside: outcome noise alone can produce a large error for an assignment whose expected MSE is modest, so squared errors are averaged over outcomes before assignments are ranked. A design with one possible assignment has no spread across assignments, but its error can still be high, and finite outcome replication adds noise to estimated risks and to which assignments appear in the worst decile.

North Carolina sudden unexpected death application

We aggregated corrected county-year SUD counts prepared by Habib[12] and Surveillance, Epidemiology, and End Results (SEER) program populations aged 18–64 into 58 community-college service-area clusters partitioning the 100 counties of North Carolina (NC). SUD deaths were identified from death certificates with the case-adjudication algorithm of Gan et al.[13] The corrected numerator includes heart-failure deaths previously excluded as presumed non-sudden: 23,523 deaths over 25,594,321 person-years during 2018–2021. Annual rates were deaths/population per 100,000. Every county mapped to one cluster; annual cluster totals reproduced statewide totals. Observed rates were held fixed with no synthetic

incidence generation (Figure 1). The four years were separate planning inputs, not longitudinal intervention follow-up.

Primary education outcomes used Equation 1 with $\beta = 0$, $\tau = 1$ and the 58-cluster row-standardized queen contiguity matrix W. Incidence directed allocation only. The fit used intercept, treatment and true spillover without incidence adjustment.

All designs were adapted before performance evaluation. Checkerboard became a deterministic graph version that alternates arms between neighboring clusters as far as the irregular map allows, not a literal checkerboard. Incidence-guided saturation used four fixed geographic regions selected from 250 k-means candidates by compactness and population/cluster/county balance, using average annual population (pooled person-years divided by four) as the reference population. Incidence-guided saturation ranked annual per-capita cluster rates, averaged cluster ranks equally within regions, then ordered regional saturations 80/60/40/20%. The grid rule averages its incidence covariate directly. The application mean-rank rule aligns with rank-scaled Poisson surfaces but differs from continuous grid incidence; it also differs from pooling deaths/population. 2×2 Blocking became spatial blocking: a random half of each of 14 groups of nearby clusters, formed once per year by k-means clustering on standardized location, incidence rank and log-population. Other strategies retained their mechanisms. SRS, Balanced Quartiles and High Incidence Focus treated 29 clusters. Rounded local budgets were retained and reported rather than equalized retrospectively.

These adaptations preserve each strategy's allocation mechanism while acknowledging the irregular graph and unequal populations. Cluster counts and working-age population shares accompany accuracy because treating half the clusters does not imply reaching half the population. The population denominator used to calculate historical rates is also distinct from the eventual sample of healthcare professionals whose education outcomes would be measured.

Primary queen settings crossed $\rho = 0, 0.2, 0.5$, $\gamma = 0.5, 0.8$, both regimes, four years, the six designs, the two variants and SRS. Sensitivities added a hypothetical rank-scaled incidence baseline together with incidence adjustment, used rook weights at $\rho \in \{0, 0.5\}$ and $\gamma \in \{0.5, 0.8\}$, or replaced incidence-guided saturation's regional mean rank with mean raw rates or pooled regional rates. In total, 1,248 settings were reported from 936 distinct

simulations: settings with the same set of possible assignments (for example, SRS in different years) shared one simulation, with annual budgets and population shares recalculated. We drew 100 assignments and simulated 100 independent outcomes for each distinct assignment, using 1,000 outcomes for designs with only one possible assignment. A prespecified rule increased replication until the MCSE of mean MSE was at most 5% of its value and the MCSE of coverage was at most 0.01; these MCSEs accounted for repeated assignments and for both assignment and outcome variation. A separate extension at $\rho \in \{0, 0.5\}$ and $\gamma = 0.8$ kept the same assignments and increased outcomes to 400 per assignment. To avoid selection bias, worst-decile assignments were identified with one half of the simulated outcomes and evaluated with the other half. Every random draw used a fixed, reproducible seed. We repeated the application for random subsets of 15 to 58 participating colleges to estimate the number of colleges needed for 80% power, using a t-based test and interval with a degrees-of-freedom-corrected standard error (Supplementary Section S11). The application used deidentified death-certificate data on deceased individuals, aggregated to 58 clusters, obtained under University of North Carolina at Chapel Hill Institutional Review Board protocol 14-2036, "Sudden Unexpected Death in North Carolina" (principal investigator Ross J. Simpson, Jr., Department of Medicine). [Placeholder for the IRB determination and consent statement, pending confirmation from Dr. Simpson.]

Results

Simulation study: the spillover regime decides which designs perform best

Table 2 reports the full queen benchmark at $\tau = 1$; Figure 2 separates the five incidence configurations, which changed each design's MSE modestly (by under 10%, except High Incidence Focus). Incidence-guided saturation's pooled mean MSE was 0.0907 against SRS 0.1285, but regime separation was essential: control-only MSE was 0.1355 against 0.2146 (ratio 0.631), whereas both-arms MSE was 0.0459 against 0.0424 (ratio 1.083). SRS and Balanced Quartiles were competitive under both arms. Coverage for designs whose effect was identified was approximately 94–95%; rook Checkerboard was not identified and was substantially biased. Rook results and effect-size/configuration breakdowns are reported separately in the supplement.

The regime difference was especially large for interspersed assignments. Checkerboard MSE increased from 0.0624 under both arms to 2.1114 control-only; 2×2 Blocking increased from 0.0623 to 0.5761. Their poorer control-only accuracy reflects imprecision rather than a comparable increase in bias. In the Poisson configuration with $\rho_X = 0.2$, empirical variance accounts for more than 97% of mean MSE for every displayed design and regime (supplementary material). Isolation Buffer behaves differently: no treated cluster has a treated neighbor, so multiplying its neighboring-treatment exposure by the control indicator leaves that exposure unchanged. Its two spillover regimes therefore define the same outcome model; small observed mean differences arise from separate Monte Carlo draws.

Pooled across both spillover regimes, the MSE ordering of the six detailed designs was stable across the five incidence configurations and five treatment-effect sizes, with Incidence-Guided Saturation lowest; SRS was not part of this ranking, and the pooled order does not imply the same order within each regime. Under control-only spillover, Isolation Buffer and Incidence-Guided Saturation were nearly level at $\tau = 2$ (MSE 0.1596 and 0.1604); at $\tau = 3$, Isolation Buffer was lower (0.1596 versus 0.1822). This crossover limits a blanket claim across effect sizes. Effect-size comparisons also reused assignments and noise, so they are sensitivity comparisons rather than independent confirmations.

The grid sensitivity omitting spillover illustrates why an apparently favorable error metric is insufficient. Among the six-design comparisons at $\tau = 1$, excluding non-identified rook Checkerboard, pooled bias ranged from approximately -0.12 to -0.40 and coverage from 0.54 to 0.91. Some designs had lower MSE under this misspecified fit because their reduced variance outweighed increased bias. That trade-off did not restore valid treatment-effect inference.

The focused grid pilot supported saturation's control-only downside pattern (Figure 3). Incidence-guided saturation's estimated worst-decile mean MSE was 0.1936 against SRS 0.3486; under both arms it was 0.0582 against 0.0544. Increasing the number of outcomes per assignment preserved the control-only direction but did not remove uncertainty about allocation tails. These are pilot values and do not replace full-study estimates.

North Carolina application: incidence-guided saturation is strong under control-only spillover

All 1,248 reported application settings completed and met the precision targets. The main run used 7,598,000 independent outcome fits; the separate extension added 2,280,000 new outcomes, for 9,878,000 distinct outcomes in total. The maximum relative MCSE of mean MSE was 0.04871 and the maximum MCSE of coverage 0.00892. There were no warnings, non-identified coefficients, estimates at the boundary of the spatial parameter space, or estimation failures. All randomized designs met the targets at 100 assignments and 100 outcomes; two baseline-sensitivity settings with a single possible assignment required 2,000 outcomes.

Under control-only spillover, incidence-guided saturation's yearly descriptive mean MSE was 21.1–29.9% lower than that of SRS (0.30761), depending on year (Figure 4; absolute annual values in the supplementary material). It had lower MSE than SRS in every primary parameter setting, and the approximate MCSE-based interval for each setting-specific difference excluded zero. High Incidence Focus's control-only accuracy varied markedly across years. Under both arms, SRS mean MSE was 0.07638 and incidence-guided saturation 0.07428–0.07682. These small differences do not establish equivalence or universal optimality. Primary yearly mean coverage was approximately 92.6–93.6%, below nominal; passing precision targets did not repair undercoverage.

The annual comparison preserves the broad regime distinction while showing changes introduced by the NC geography. On the NC map, Checkerboard is identified, yet has roughly three times SRS MSE in both regimes. Isolation Buffer is close to SRS control-only but has approximately 3.74 times its both-arms MSE. 2×2 Blocking (spatial blocking in NC) is close to SRS under both arms and more variable across planning years control-only (Figure 4). Thus transferring a grid mechanism does not transfer its numerical ranking unchanged. These descriptive averages are conditional on fixed observed incidence and the adapted assignment rules.

At $\rho \in \{0, 0.5\}$ and $\gamma = 0.8$ (queen weights), control-only incidence-guided saturation had estimated q90 20.8–38.6% below SRS and worst-decile mean 20.7–43.4% below. The pattern held when one half of the outcomes selected the worst decile and the other half evaluated it

(detailed displays in the supplementary material). However, only ten sampled assignments formed the worst decile, and all 112 reported settings with more than one possible assignment had uncertain membership at the worst-decile boundary. Under both arms, the worst-decile sets selected by the two outcome halves for incidence-guided saturation overlapped little (Jaccard index 0–0.25), and about 61–86% of the variance in estimated risk was attributable to Monte Carlo error from the finite number of simulated outcomes. Small both-arms tail differences cannot support superior downside protection. Checkerboard and High Incidence Focus each have a single possible assignment for these inputs, so no allocation variation, but can still have substantial estimation error. Among the other designs (supplementary material), Isolation Buffer also had a lower control-only worst-decile mean than SRS (0.62 of SRS in the grid pilot and 0.91 in the application) but 3.7–3.8 times SRS mean MSE under both arms. High Incidence Focus's fixed application allocation was below the SRS worst-decile mean in both regimes (0.64 control-only, 0.76 both arms), but its grid-pilot both-arms mean was 3.91 times that of SRS.

Under both-arms spillover, every design except Checkerboard and Isolation Buffer needed 35 to 37 colleges for 80% power at a one-standard-deviation effect. Under control-only spillover, no design reached 80% power at that effect with all 58 colleges, and the saturation designs' advantage over SRS fell from 22% lower MSE with 58 colleges to 4–5% with 29.

Budgets differed (annual cluster/population tables and displays in the supplementary material). Checkerboard treated 27 clusters under queen; Isolation Buffer averaged 15.89; 2×2 Blocking treated 28/31/27/29 by year, including a one-cluster group always assigned control under the retained rounding rule. High Incidence Focus treated 29 clusters but approximately 22–24% of working-age population; Isolation Buffer reached about 27%. These shares describe geographic reach, not healthcare-professional enrollment or cost. On the NC map, Checkerboard's effect was identified but imprecise, unlike its exact non-identification under rook weights on the rectangular grid.

A sensitivity analysis that added a hypothetical incidence-related baseline to the outcome and adjusted for incidence in the fit increased High Incidence Focus MSE roughly 3.68–4.44-fold under both arms and 1.42–2.52-fold control-only, across settings. Aligning treatment with incidence left less information about τ after adjustment; baseline and adjustment changed

jointly. Rook and queen results at matched parameter values led to the same broad conclusions. Alternative regional summaries changed some region orders, with no retrospective replacement of the prespecified mean-rank rule.

Discussion

Implications for trial design

The completed grid and observed-incidence application support conditional guidance: consider saturation when spillover primarily reaches controls, and retain SRS among competitive options when both arms receive exposure. Average MSE and allocation downside should be evaluated together. A restricted design's near-zero allocation variability is insufficient if its conditional error is high. Conversely, small estimated tail differences with noisy membership are insufficient to dismiss SRS. Isolation Buffer's control-only downside protection is useful only when both-arms spillover is implausible and fewer treated clusters are acceptable. Incidence can also be used to target treatment directly. High Incidence Focus illustrates the trade-off: low conditional both-arms error in the primary education model came from a fixed assignment, not a demonstration of randomized causal identification. Its sensitivity to incidence adjustment also prevents a general recommendation based on that result alone. For trial planning, the combined findings favor evaluating plausible exposure regimes before selecting an allocation. Saturation creates variation in local treatment proportions while retaining treatment and control clusters; its favorable control-only accuracy is consistent with the need to distinguish direct treatment from neighboring exposure. Under both-arms spillover, SRS and incidence-balanced assignments already provide competitive estimation, making a modest mean difference an insufficient reason to impose a more restricted design.

Limitations

These comparisons complement spatial analysis rather than replace it. The primary fit knows the spillover mechanism; the grid sensitivity omitting it showed bias and poor coverage. We did not establish performance with incorrect exposure networks, heterogeneous treatment effects or participant-level sampling error. The grid and one irregular geography do not span all trial structures, and unequal treatment budgets prevent interpreting the rankings as

equal-cost optimization. Historical incidence was fixed; measurement uncertainty and change between planning and implementation were not propagated.

Wald intervals under-covered; the t-based interval restored coverage to 94–95% on the grid and 94–96% in the application at every number of participating colleges (Supplementary Section S11). Worst-decile estimates are conditional on the sampled assignments and finite outcome replication; the extension improved risk estimates for the same assignments rather than drawing new ones. Results shared across years also cannot be treated as independent evidence.

Continued work for the education pilot

A practical decision should also compare recruitment and exposure consequences alongside MSE. Isolation Buffer's smaller treated-cluster budget and High Incidence Focus's smaller population reach mean that these results do not answer which design is most efficient for a fixed implementation cost or participant sample. Before using them to choose one pilot allocation, investigators would need to define the education endpoint, identify eligible professionals, specify cluster and participant budgets, and assess whether geographic contiguity represents actual communication. Those choices could alter outcome precision or the exposure network.

Conclusions

The education application is a completed design evaluation, not an implemented intervention or evidence that education reduces SUD. Within the evaluated assumptions, saturation offers favorable average and descriptive downside performance under control-only spillover; SRS remains a reasonable comparator and candidate under both-arms exposure.

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Author contributions

AW developed the method, designed the simulation study, implemented the application study, prepared all figures and wrote the manuscript. AH prepared the North Carolina SUD datasets for the application study. RS proposed the SUD prevention education pilot. FL provided methodological guidance and review. All authors critically revised the manuscript and approved the final version. [Placeholder: pending coauthor approval.]

Declaration of conflicting interests

The authors declare that there is no conflict of interest.

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Data availability

All data supporting the findings of the simulation study were generated by simulation. The simulation data and R code are available from the authors on request. The North Carolina death-certificate data used in the application are restricted and are not publicly available.

Supplementary material

The supplementary material gives full specifications of all designs and their North Carolina adaptations, the complete simulation parameter grid and estimation diagnostics, and detailed simulation results, including rook contiguity, analyses omitting the spillover term and results by treatment-effect size. For the North Carolina application, it adds cluster maps, results and cluster treatment allocations for each planning year, and the sensitivity analyses.

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Table 1. Treatment assignment designs on the 100-cluster grid and their North Carolina (NC) adaptations, each with its category (italic) and one example assignment on the grid (blue, treated; white, control). Every example uses the illustrative incidence surface in the first row, which is not one of the simulated surfaces. Numbers in the Balanced Quartiles example give each cluster's incidence quartile; percentages in the Incidence-Guided Saturation example give each quadrant's treated share. Incidence is the simulated grid covariate or the observed annual cluster rate. Two related variants, saturation with randomly ordered quadrants and balancing within incidence halves, performed similarly to their retained counterparts and are reported in the supplementary material.

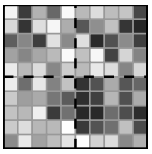
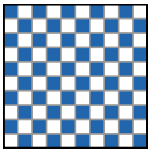
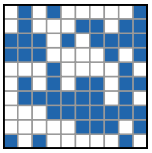
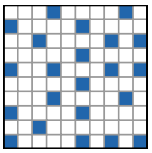
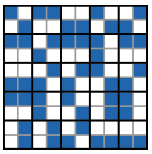
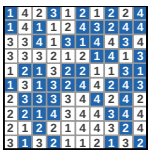
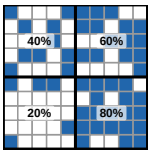
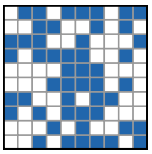
Design	Example (grid)	Grid (100 clusters)	NC (58 clusters)
Incidence surface		Illustrative surface used in every example below; darker is higher, and dashed lines mark the quadrants.	Observed annual SUD rate of each cluster.
Checkerboard *Fixed spatial interspersion*		Treat alternating clusters in a checkerboard pattern, so no two clusters that share an edge are in the same arm (50 treated).	Deterministic graph version: visit clusters from most to fewest neighbors and give each the arm less common among its assigned neighbors (27 treated).
High Incidence Focus *Incidence targeting*		Treat the 50 clusters with the highest incidence; ties are broken at random.	Treat the 29 clusters with the highest observed rate.
Isolation Buffer *Spatial separation*		Treat clusters one at a time, each chosen at random from the untreated clusters with no treated neighbor, until none remain; no two treated clusters are neighbors (about 21 treated under queen weights).	Same rule on the NC adjacency graph (about 16 treated).
2×2 Blocking *Local spatial balance*		Divide the grid into 25 blocks of 2×2 clusters and treat two randomly chosen clusters in each block (50 treated).	Treat a random half (rounded) of each of 14 fixed groups formed by k-means clustering on location, incidence rank and population (27–31 treated by year).
Balanced Quartiles *Incidence balance*		Divide clusters into quartiles of incidence rank and treat a random half of each quartile (50 treated).	Same rule with observed rates (29 treated).
Incidence-Guided Saturation *Incidence-ordered regional saturation*		Treat a random 80%, 60%, 40% and 20% of the clusters in the four 5×5 quadrants, ordered from highest to lowest mean incidence (50 treated).	Same rule with four fixed geographic regions, ordered by the mean rank of their clusters' observed rates (29 treated).
Simple Random Sampling *Unrestricted reference*		Treat a completely random half of the clusters (50 treated).	Same rule (29 treated).

Table 2. Grid benchmark at $\tau = 1$, queen contiguity. Each regime-specific entry averages 80 scenarios (five incidence configurations, four ρ and four γ values). MSE is the mean squared error of the estimate of τ; coverage is the proportion of 95% t-based confidence intervals, with a degrees-of-freedom-corrected standard error, that contain τ. Both use the oracle fit, the spatial lag model that includes the true spillover term. Simple random sampling is complete randomization of 50 clusters.

	Both arms		Control only	
Design	MSE	Coverage	MSE	Coverage
Checkerboard	0.0624	0.948	2.1114	0.951
High Incidence Focus	0.1617	0.950	0.3292	0.945
Isolation Buffer	0.1608	0.948	0.1595	0.950
2×2 Blocking	0.0623	0.947	0.5761	0.950
Balanced Quartiles	0.0421	0.949	0.2195	0.942
Incidence-Guided Saturation	0.0459	0.949	0.1355	0.945
Simple Random Sampling	0.0424	0.950	0.2146	0.946

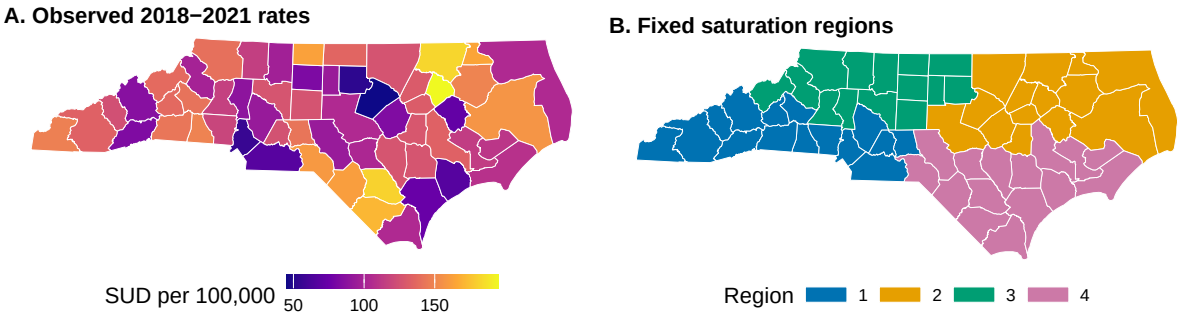


Figure 1. Observed North Carolina planning inputs. (A) Observed SUD rates per 100,000 population aged 18–64 for the 58 service-area clusters, pooled over 2018–2021 (deaths divided by person-years); the analysis evaluates each year separately. (B) The fixed four-region partition used by incidence-guided saturation in every year. Incidence guides primary allocation only; it does not enter the education-outcome baseline. All-year maps and assignment examples are in the supplementary material.

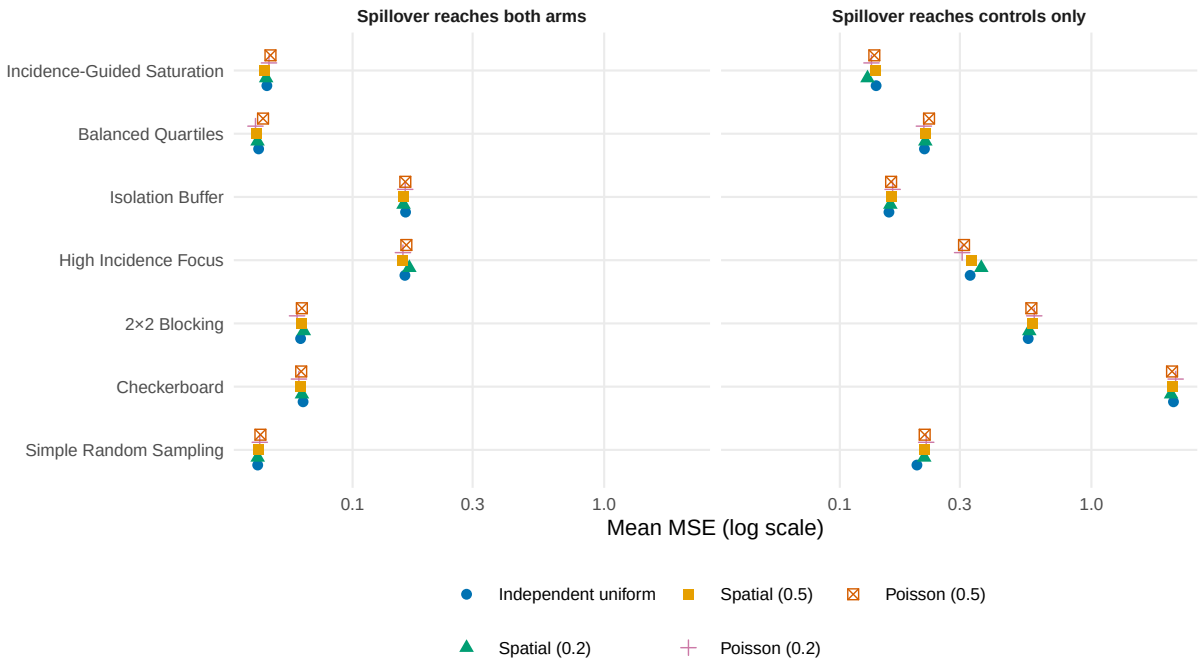


Figure 2. Grid mean MSE at $\tau = 1$ with queen weights, for the six designs and simple random sampling. Points keep the five incidence configurations separate, averaging 16 outcome-dependence/spillover-magnitude settings within each configuration and regime. Parentheses in the legend give incidence dependence ρ_X. Both panels use the same logarithmic scale. Parameter-specific displays remain in the supplementary material; no pooled Monte Carlo intervals are implied.

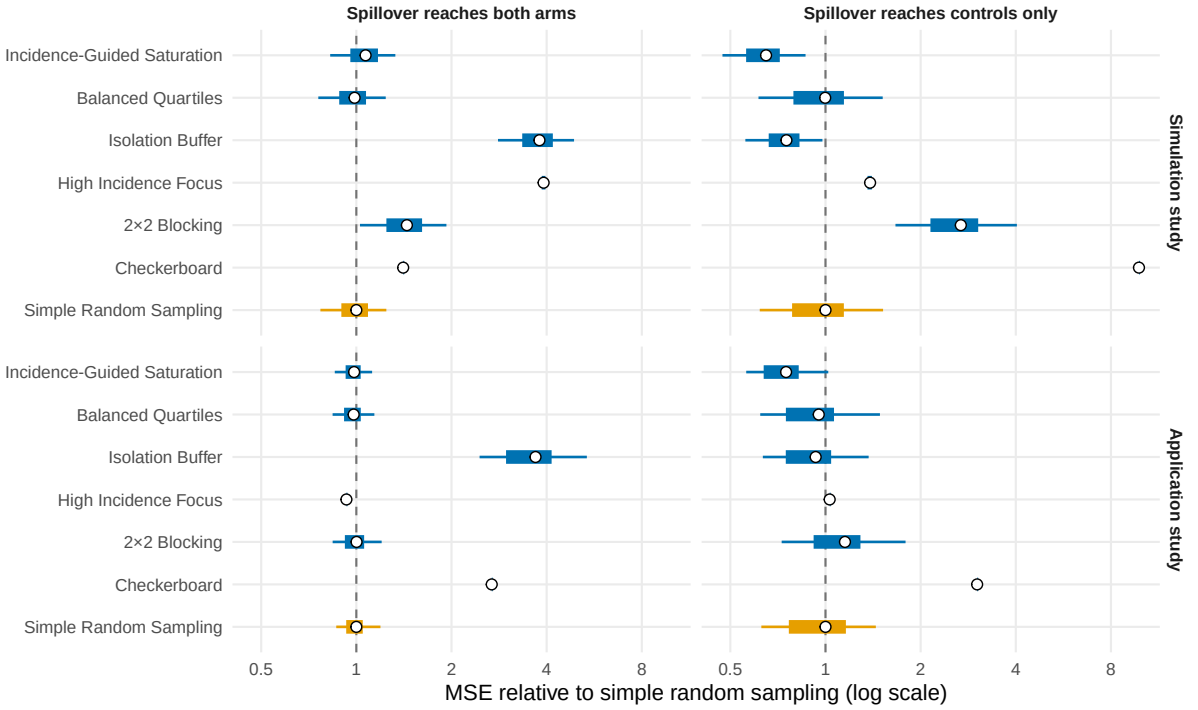


Figure 3. Distribution of allocation-specific MSE for the six designs and simple random sampling (SRS), relative to the mean MSE of SRS in the same setting (queen weights). Simulation study: focused grid pilot at $\tau = 1$, 40 fixed surface/parameter settings per spillover regime. Application study: NC settings with extended outcome replication ($\rho = 0, 0.5; \gamma = 0.8$) in each year 2018–2021. For each allocation, MSE is estimated from repeated simulated outcomes (100 in the simulation study, at least 400 in the application study). Thin lines span the 5th to 95th percentiles of allocations and thick bars the 25th to 75th; circles mark the mean of the setting-specific ratios. SRS is shown in orange and the designs in blue. Percentiles are computed within each setting and then averaged, so bars show allocation-to-allocation spread; a design with one allocation per setting appears as a circle only. Monte Carlo error from the finite number of outcomes per allocation widens the bars slightly.

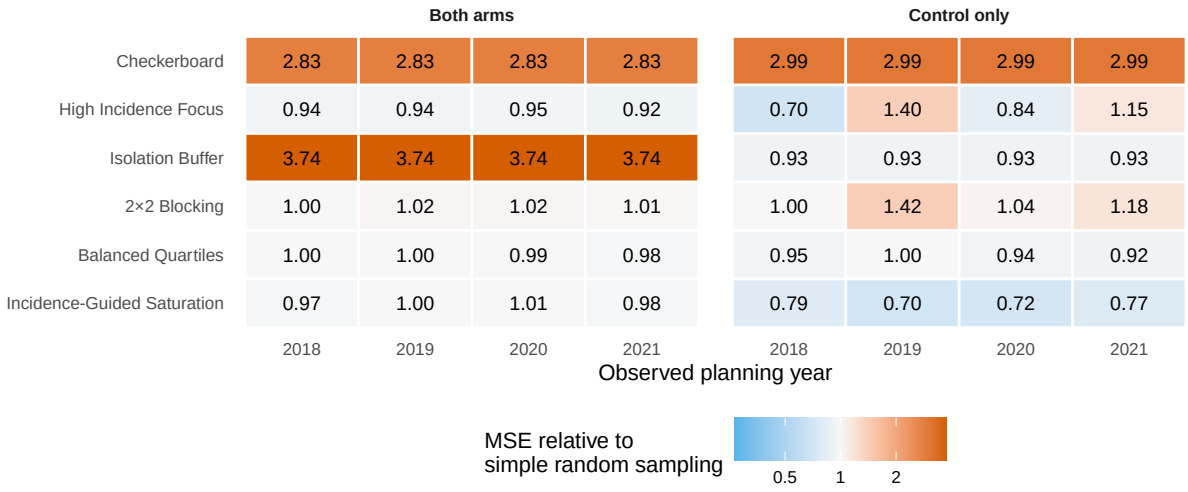


Figure 4. Completed primary application: yearly mean MSE divided by year- and regime-matched simple random sampling (SRS) MSE, for the six designs with queen weights and education outcomes. Each cell averages six ρ/γ settings equally before taking the ratio. Blue indicates lower error than SRS, orange higher error; colors use a logarithmic ratio scale and cells print the ratios. SRS mean MSE is 0.07638 under both arms and 0.30761 control-only. Years that share one simulation (designs whose assignments do not use incidence) are not independent replications. Absolute MSE, bias, coverage, budgets and setting-specific comparisons are in the supplementary material.